

What is an open platform?

Khasim Syed Mohammed explains, "An open platform is where hardware and software for a platform are made available without any further constraints and restrictions. The user or consumer is allowed to design, develop, validate, maintain and enhance the feature set of a product. This requires a committed team work to become successful."

As the hardware and software are developed in an open environment, both play an important role in defining the end platform. The licensing terms and conditions on software and hardware components should be adaptable without any restriction on usage, distributions and improvements.

of a boot loader, various libraries, Linux kernel and applications running on top of it.

What's in a board?

The revision C of Beagle Board was introduced in May 2009—in India, Cranes Software International is the official distributor. So, coming back to our real question: What can a developer look for in the board?

First, the OMAP3530 processor. OMAP3530 is a high-end, multicore application processor with ARM Cortex A8, Generic C64x DSP and an SGX 3D graphics accelerator. With the ARM running at 600 MHz, DSP at 430 MHz and SGX at 111 MHz, the effective total system power becomes 1.2 GHz. This gives developers an opportunity to explore a variety of desktop applications on handheld devices with much lower power consumption than any other x86- or Intel Atom-based device. What's even more interesting is that the board can run desktop applications—the OpenOffice.org office suite, GIMP, Firefox, etc—with only 2 watts of power consumption. With respect to heat dissipation, the OMAP3530 processor is rather superior. There's no need for any sort of cooling device (like a heat-sink or a fan) as the temperature remains below 50 degrees Celsius. Now, that's what we call an ideal green computing device.

Although there are a lot of interfaces available on the board, the high-speed USB and USB OTG are worth mentioning first. A single high-speed USB on the board can be used to connect 128 devices with the help of a hub. However, when we use a high-speed USB port, the USB host acts only as the

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master and the device acts as a slave. A USB OTG allows the host to behave either as a master or a slave, but not at the same time. When the USB OTG is behaving as a host, it should be connected to a 5V power supply. But, when connected to a PC, it behaves as a slave.

It's no surprise that the device, according to the manufacturers, is aimed at exploring and developing applications on OpenGL, OpenMax, OpenOffice.org, Open CV, Browsers, 3D gaming, GStreamer, 3D UI, etc. But would the device be able to run all these applications in parallel? In which case, how much power will the device typically consume? Would a cellphone-like battery be enough to support it for hours? Also, how much RAM can be fitted into the board to take care of intensive multi-tasking?

Syed quips, "The RAM is fixed for 256 MB and is not expandable; therefore memory requirement is a constraint to run all these applications in parallel. But it's a board limitation rather than one related to the processor. A cellphone-like battery is sufficient to support all these applications, but definitely not for hours when run in parallel. The OMAP3530 chip supports about 82 hours of audio

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